
A novel tactile tomography system based on mechanical principles for internal 3D imaging

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Mechanical principle-based imaging technologies, such as contact profilometers, atomic force microscope (AFM), and AFM-derived imaging modes, have been widely utilized in industrial detection and materials characterization. However, these imaging technologies primarily focus on detecting surface profiles and morphology, lacking the capability for internal 3D imaging. Herein, a novel tactile tomography system based on mechanical principle is proposed to image the internal structure of an object. This tactile tomography system comprises a scanning module, a probe module, and a closed-loop feedback control system, which enables it to recognize the softness of soft matter. Then a novel parameter of “tactile softness of materials” was constructed by simulating human tactile perception of softness. By considering the combined effects of the sample’s size and thickness, and the applied force on the tactile softness of materials, the tactile tomography system achieves internal 3D imaging capability with a maximum detection depth of 6 mm. The scanning resolution of this tactile tomography system is as high as 0.5 mm in both the x and y directions, and 0.1 μm in the z direction. The tactile tomography system has a scanning accuracy exceeding 98.9%, and an optimal scanning speed and scanning step of 2.4 mm/s & 0.5 mm. Successful application in defect detection of encapsulated flexible printed circuit boards (FPCBs) demonstrated the promising potential of the tactile tomography system, marking a significant advancement in internal 3D imaging technologies based on mechanical principles.

1. Introduction

Imaging technologies play a crucial role in medical imaging, industrial detection, materials characterization in physical science, and geological detection in earth science [1–4]. Based on imaging principles, these imaging technologies can be categorized into optical imaging, X-ray imaging, ultrasonic imaging, nuclear imaging, magnetic resonance imaging (MRI), electron microscope, and mechanical imaging [5–11]. The primary functions of these imaging technologies include visualizing the surface morphology, profile, or internal structure of the measured sample [12–14]. Especially the imaging of internal structure – achieved by technologies such as X-ray computed tomography (X-CT), MRI, ultrasonic tomography (U-CT), optical computed tomography (O-CT), and positron emission tomography (PET), has significantly advanced the fields of medical imaging, industrial detection, and geological detection [15–19]. Consequently, internal 3D imaging is of great importance for imaging technologies. However, imaging technologies based on mechanical principles still lack the capability for internal 3D imaging.

Imaging technologies based on mechanical principles include contact profilometers and atomic force microscope (AFM), and AFM-derived imaging modes [20–22]. Contact profilometers are typically classified into two categories: stylus profilometers and three-coordinates measuring machines (CMM). The stylus profilometer excels at detecting surface roughness and morphology of a workpiece, while the CMM is

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Fig. 1. Schematic diagram of the tactile tomography system.

[block diagram: Computer $\leftrightarrow$ XYZ-axes position signals $\leftrightarrow$ Scanner mode; a Programmable Logic Controller (PLC) drives the Z stage (with grating ruler, probe, pressure sensor above the sample) via Z-axes driving signals and receives Z-axes position signals; PLC also drives the XY stage via XY-axes driving signals; a Data collection and processing cell sits between the PLC, the Z stage, and the sample]

Fig. 2. Photograph of the overall tactile tomography system.

[photograph with callouts: Computer (monitor + desktop on the left), Data collection and processing cell, Z-stage, X-stage, Y-stage, Probe, Grating sensor, PLC (cabinet at the bottom)]

more adept at obtaining 3D surface morphology, offering advantages such as high precision, good repeatability, and ease of operation [23,24]. However, as both are restricted to surface profiling, they cannot penetrate beneath the surface or resolve internal structures, limiting their application to external feature analysis.

AFM equipped with a micro-cantilever is highly sensitive to weak force. One end of the micro-cantilever is fitted with a tip. Surface imaging of AFM relies on the interactive forces between the tip and the sample, and it can operate in three basic modes: contact mode, non-contact mode, and tapping mode [25]. The interactive forces may include Van Der Waals forces, electrostatic forces, or magnetic forces, which cause the micro-cantilever to deform or vibrate. This deformation or vibration can be used to infer the surface morphology and physical properties of the sample. Currently, AFM can provide detailed information about surface morphology, roughness, height, thickness, and phase diagrams with a high resolution of 5 nm [26]. Consequently, AFM has been widely utilized in materials science, surface science, biomedicine, and other fields. In addition, AFM has developed several derivative modes, including kelvin probe force microscopy (KPFM), piezoresponse force microscopy (PFM), peak force quantitative nano-mechanics mapping (PF-QNM), conducting atomic force microscopy (C-AFM), and lateral force microscopy (LFM) [27–31]. Beyond detecting surface morphology, these derived imaging modes offer unique functionalities. For instance, KPFM can measure work function and surface potential [27], PFM can provide information about electric domains [28], and PF-QNM can yield data on Young’s modulus, adhesive force, and maximum deformation [29]. Although these AFM-derived imaging modes contribute to the abundant functions of the mechanical principle-based imaging technologies [30,31], they suffer from critical limitations for internal imaging. They rely on surface interactions with a penetration depth of less than 100 nm, limiting

Fig. 3. Mechanical design of the scanning module and probe module.

[CAD rendering: on the left, a cross-shaped XY sliding table with labels “X-axes linear sliding table module,” “Y-axes linear sliding table module,” and “Sample” on the central platform; above the cross, a vertical “Z-axes rodless cylinder” with a “Grating sensor” and “Probe” mounted to its underside. On the right, an enlarged detail of the probe: a narrow tube of total length ≈ 150 mm, tip diameter 0.5 mm, terminating in a “Pressure sensor” module]

Fig. 4. Flow chart of the closed-loop feedback control system.

[main flow on the left: Planning scan path $\rightarrow$ PLC $\rightarrow$ “The X and Y axes move” $\rightarrow$ “Z-axis descent” $\rightarrow$ decision diamond “The pressure sensor reaches the first threshold” (NO loops back to Z-axis descent, YES feeds the grating-sensor position reader) $\rightarrow$ “The pressure sensor reaches the second threshold” $\rightarrow \dots \rightarrow$ “The pressure sensor reaches the final threshold” $\rightarrow$ “Z-axis stops and resets.” A left-hand feedback loop $((Z_0, Z_1, \dots, Z_n))$ carries position data from the grating sensor back up into the PLC. On the right, a visual strip shows the probe descending through layers of a stratified soft sample at successive depths $Z_0, Z_1, \dots, Z_n$.]

Fig. 5. Softness recognition of the tactile tomography system.

- (a) 3×3 grid of nine AB glue samples, labeled with Shore hardnesses 3, 7.5, 12.5, 17.5, 25, 30, 35, 50, and 60 HC.
- (b) Shore hardness tester (LX-C), photograph of the bench-top analog dial instrument with inset views of LX-A, LX-C, and LX-D indenter tips.
- (c) Compression depth distribution over the 3×3 AB glue grid as scanned by the tactile tomography system. X, Y in mm; Z color scale 0 (dark blue) to 6 mm (dark red), showing compression depth decreasing as Shore hardness increases.
- (d) Relation between compression depth and Shore hardness. Measurement points (black squares) with a quadratic fit (red curve): $Y = 6.23 - 1.62 X^1 + 1.15 X^2$, $R^2 = 0.99596$. Compression depth axis: 0–7 mm; Shore hardness axis: 0–70 HC.]

Fig. 5. Softness recognition of the tactile tomography system. (a) AB glues with Shore hardnesses of 3, 7.5, 12.5, 17.5, 25, 30, 35, 50, and 60 HC. (b) Shore hardness tester (LX-C). (c) Compression depth distribution when the tactile tomography system scanned the AB glue samples with different Shore hardnesses (scanned by the tactile tomography system). (d) Relation between compression depth and Shore hardness.

Fig. 6. Effect of the size of the sample on the tactile softness of materials.

- (a) Frame model with a depth of 6 mm and different length & width (3D printed ABS grid).
- (b) Frame model filled with a soft surface layer of AB glue. Labeled scanning area 8×8 mm, with decreasing lengths 8.0, 6.4, 5.1, 4.0, 3.2 mm marked along each axis.
- (c) 3D imaging of compression depth distribution when the tactile tomography system scanned the frame model with a soft surface layer. X, Y in mm; Z 0–6 mm (color bar).
- (d) Top view of (c), showing that the central cells of the grid have the largest compression depth and that depth decreases as cell size shrinks.]

Fig. 6. Effect of the size of the sample on the tactile softness of materials. (a) Frame model with a depth of 6 mm and different length & width. (b) Frame model filled with a soft surface layer of AB glue. (c) 3D imaging of compression depth distribution when the tactile tomography system scanned the frame model with a soft surface layer. (d) Top view of the 3D imaging of compression depth distribution when the tactile tomography system scanned the frame model with a soft surface layer.

analysis to the outermost surface or near-surface layers. Moreover, they cannot handle thick samples ($> 1 \mu\text{m}$) or soft-encapsulated materials (such as flexible electronics with soft coatings), as probe-sample interactions are too weak to penetrate without damaging the surface.

Recent advancements in mechanical principle-based tomography have begun to address the gap in internal imaging, with Mechanics Based Tomography (MBT) emerging as a promising approach [32–34]. MBT leverages solid mechanics equations and surface displacement measurements to reconstruct internal material property distributions through an optimization framework with regularization to handle ill-posed inverse problems. Experimental validations on composite silicone samples have demonstrated MBT’s ability to visualize subsurface inclusions by resolving stiffness contrasts, highlighting its potential for non-destructive testing of soft materials [34]. Unlike X-CT or ultrasonic methods, mechanical principle-based tomography avoids ionizing radiation or high-energy waves, making it suitable for delicate samples. In addition, mechanical principle-based tomography can map mechanical properties (such as stiffness or elastic modulus), providing functional information beyond structural imaging. However, existing mechanical tomography methods face challenges with low reconstruction accuracy and reliability, as displacement measurement noise and position deviations of the force can both lead to a significant deterioration in the quality of internal structure reconstruction.

In this study, we propose a novel tactile tomography system based on mechanical principles, featuring internal 3D imaging capabilities. The tactile tomography system is constructed from a scanning module, a probe module, and a closed-loop feedback control system, enabling it to sense force and control the movement of the probe tip. This tactile tomography system can further recognize the softness of soft matter. To simulate human tactile perception of softness, a novel parameter of “tactile softness of materials” is proposed. The tactile tomography system can achieve the function of internal 3D imaging with a maximum detection depth of 6 mm by the combined effects of the sample’s size, thickness, and the applied force. The performance of the tactile tomography

system is characterized by its scanning resolution, scanning accuracy, and the trade-off between scanning efficiency and imaging quality. Finally, the promising potential of the tactile tomography system is illustrated through its application in defect detection of encapsulated FPCBs.

2. System design and instrumentation

The configuration of the tactile tomography system is shown in Fig. 1, and the photograph of the developed tactile tomography system is presented in Fig. 2. The tactile tomography system consists of a scanning module, a probe module, and a closed-loop feedback control system.

2.1. Scanning module

The mechanical design of the scanning module and probe module is shown in Fig. 3. This design incorporates two linear actuators (MBDLN25SE, Panasonic) and two servo motors (MSMF022L1U2M, Panasonic) that together form an XY stage, allowing for precise adjustments of the measurement position on the sample. A rodless cylinder (CDHD-4D52AAP1) is utilized to drive the probe module, applying pressure to the sample in the vertical direction, and a grating sensor (RXF1188508-01) measures the depth of the applied pressure. The movement speeds of the XY stage and Z stage can be adjusted from 0.1 to 500 mm/min and from 0.1 to 400 mm/min, respectively. The XY stage and Z stage achieve a limit resolution of step displacement of 10 μm and

Fig. 7. Effect of the thickness of the sample on the tactile softness of materials.

- (a) Hierarchical structure model with 9 staircase steps, 0.5 mm height difference between adjacent stages (total run 5 mm), CAD view with the stepped top surface.
- (b) Hierarchical model filled with a soft surface layer of AB glue (photograph), uniform square scanning region outlined.
- (c) 3D surface plot of compression-depth distribution when the tactile tomography system scanned (b); stepped pyramid shape with X, Y in mm and Z 0–6 mm on a rainbow color scale.
- (d) Bar chart: measured height (mm, 0–7 axis) at step numbers 1–9; consecutive step-to-step height differences are 0.494 mm and 0.495 mm, matching the 0.5 mm nominal staircase step.]

Fig. 7. Effect of the thickness of the sample on the tactile softness of materials. (a) Hierarchical structure model with the height difference between each adjacent stage of 0.5 mm. (b) Hierarchical structure model filled with a soft surface layer of AB glue. (c) 3D imaging of compression depth distribution when the tactile tomography system scanned the hierarchical structure model with a soft surface layer. (d) Height of the compression depth distribution.

0.1 μm , respectively, enabling the tactile tomography system to operate with high scanning resolution.

2.2. Probe module

As shown in Figs. 2 and 3, the probe module of the tactile tomography system was assembled from a shell structure, a tip, two springs, and a carbon fiber beams based pressure sensor [35]. To achieve high scanning resolution without penetrating the soft layer of the sample, the tip was designed with a diameter of 0.5 mm. The springs are positioned behind the tip to transmit the force from the tip to the pressure sensor and to restore the tip's position. The pressure sensor, which is based on carbon fiber beams, is capable of quantitatively measuring force and responds to the force exerted by the springs. Consequently, the probe module can accurately capture force information when the tip makes contact with the sample.

2.3. Closed-loop feedback control system

A programmable logic controller (PLC) (H2U-48MT, Henschman) was integrated with the scanning module and the probe module to establish a closed-loop feedback control system. The workflow of the tactile tomography system is shown in Fig. 4. Firstly, scanning paths, scanning speed, and scanning step were programmed. The PLC control the X and Y axes to drive the sample to move until the target scanning point is right under the tip, and then the rodless cylinder of the Z axis drive the probe down. The resistance value of the pressure sensor changes when the tip of the probe module touches the surface of the sample. This resistance value is collected by an ADS1247 acquisition chip, which converts the analog signal into a digital signal. The digital signal is then transmitted to an STM32F030C8T6 microprocessor, where it is identified as the first threshold. Meanwhile, the position information of Z_0 is read by the grating sensor, and the microprocessor sends this position information to the PLC. The rodless cylinder then continues to drive the probe downward, while the grating sensor reads a series of positional data as the resistance value of the

pressure sensor reaches specific characteristic thresholds. The PLC stops and resets the probe once the resistance value of the pressure sensor attains the target threshold. The X and Y axes then drive the sample to the next scanning point until all preset areas have been scanned. The positional information (x, y, z) corresponding to each threshold is saved as projection data to reconstruct a 3D image of the internal structure of the sample. This projection data is stored in a CSV file and displayed in real time through a visual interface.

3. Imaging principle of tactile tomography

3.1. Softness recognition of tactile tomography system

The ability of a material sample to resist the penetration of hard objects into its surface is known as hardness, which is an intrinsic property of the material. [36]. According to different measurement approach and standard, hardness can be subdivided into Rockwell hardness, Vickers hardness, Shore hardness, and Brinell hardness. Among these hardness measurements, Shore hardness has been widely applied to the hardness testing of soft materials, such as elastomers, rubber, soft plastics, and foam [37]. Therefore, the softness of soft matter can be characterized by Shore hardness, which exhibits a negative correlation with softness. The tactile tomography system can

Fig. 8. Effect of the applied force on the tactile softness of materials.

3D imaging of compression-depth distribution when the tactile tomography system scanned the hierarchical structure model (Fig. 7) with a soft surface layer under a pressure of (a) 0.5 MPa, (b) 1.5 MPa, (c) 2.5 MPa, and (d) 4 MPa respectively. Each panel is a rainbow-colored rendered 3D surface (Z scale 0–6 mm). At low force (0.5 MPa) only the uppermost staircase plateau resolves; as force increases more layers of the stepped interior become visible, until at 4 MPa the full staircase structure is recovered.]

Fig. 8. Effect of the applied force on the tactile softness of materials. 3D imaging of compression depth distribution when the tactile tomography system scanned the hierarchical structure model with a soft surface layer under a pressure of (a) 0.5, (b) 1.5, (c) 2.5, and (d) 4 MPa, respectively.

Recognize softness by applying a constant force through a tip pressing against the soft material. This capability is attributed to the closed-loop feedback control system. As shown in Fig. 5a, a series of two-component epoxy adhesives (AB glues) with Shore hardnesses of 3, 7.5, 12.5, 17.5, 25, 30, 35, 50, and 60 HC were prepared with a size of $15 \times 15 \text{ mm}^2$ and a thickness of 6 mm, and their hardness values were measured by a Shore hardness tester (LX-C) (Fig. 5b). The tactile tomography system pressed each AB glue sample with a constant pressure of 2.5 MPa. The scanning point for each AB glue is 12, within a scanning area of $12 \times 12 \text{ mm}^2$. The scanning results are shown in Fig. 5c, where the AB glues with different Shore hardnesses exhibit different compression depths. The average compression depth decreases as the Shore hardness of the AB glues increases (as shown in Fig. 5d), demonstrating a negative correlation. This indicates that the softness of soft materials can be characterized by the compression depth. Therefore, based on the tactile tomography system, the relationship between the softness and compression depth can be defined as

$$\text{Softness} = \frac{\Delta d}{P} = \frac{\Delta d \cdot S}{F} \quad (1)$$

where Δd is the compression depth, F is the force when the tip pressing the sample, and S is the contact area between the tip and the sample.

3.2. Tactile softness of materials

The size and thickness of the sample must be taken into account when testing Shore hardness. The contact position of the specimen should be at least 12 mm from the edge (for Shore A, B, C, and D), and the thickness must not be less than 6 mm. Consequently, the compression depth will vary if the size and thickness of the sample are less than 12 mm and 6 mm, respectively. Similar to tactile softness [38], which is perceived through the mechanoreceptors in human skin, the tactile softness of materials is proposed for use in a tactile tomography system to simulate human haptic perception to recognize the softness of soft materials, taking into consideration factors such as sample size, sample thickness, and the applied force. As shown in Fig. 6a, a frame model was fabricated by 3D printing with acrylonitrile butadiene styrene (ABS). The AB glue samples were filled in the frame model to

form a series of samples with different sizes and the same thickness of 6 mm, in which the length and width were decreased from 8.0 mm to 3.2 mm (8.0, 6.4, 5.1, 4.0, and 3.2 mm, respectively) (Fig. 6b). Subsequently, the samples were scanned by the tactile tomography system under a constant pressure of 4 MPa. Fig. 6c presents that the sample with a length & width of 8 mm & 8 mm possesses the largest compression depth, and the compression depth decreases with decreasing length or width of the sample. The compression depth of the central area of each sample are larger than that of the edge area (Fig. 6d). These results demonstrate that the tactile tomography system can effectively simulate human haptic perception, allowing for the differentiation of tactile softness distribution in materials when the size of the soft matter is less than 12 mm.

In addition, a hierarchical structure model was fabricated by 3D printing with ABS, and the height difference between each adjacent stage is about 0.5 mm (as shown in Fig. 7a). Then the AB glue was filled in the hierarchical structure model to form a soft surface layer with different thickness (Fig. 7b). The image reconstructed by the projection data from the tactile tomography system scanning this sample with a constant pressure of 4 MPa is shown in Fig. 7c, the compression depth of the samples with different thickness surface layers exhibits clear gradation. The compression depth difference between each adjacent thickness is also about 0.5 mm (Fig. 7d), which consistent with the height difference of each adjacent stage. Significantly, by combining the compression depth and its corresponding location information (x and y), Fig. 7c also shows the internal hierarchical structure of the sample, indicating that the tactile tomography system can obtain the internal structure information of the sample with a soft surface layer based on tactile softness of materials.

The object that is buried deep cannot be sensed by the fingers unless a large force is applied. Similarly, the tactile tomography system is

Fig. 9. Theoretical validity of the tactile tomography.

- (a) Symbol “+” model: CAD drawing of a 50 mm × 50 mm substrate with a raised plus-sign feature at the center.
- (b) Symbol “+” model filled with AB glues: photograph showing the glue-covered sample, 50 mm × 50 mm, 20 mm glue depth marked.
- (c) Simulation results of the mechanical process of the probe pressing the sample: meshed FEA view with the probe tip (small yellow sphere) contacting the red sample block.
- (d) Simulation results of the sample scanned by the probe tip with 676 scanning points: rainbow-colored contour in the central square region, red-to-yellow elsewhere.
- (e) Simulated displacement field of the sample under probe loading: rendered 3D surface plot showing the plus-sign structure emerging via displacement contrast.
- (f) Experimental result of the sample scanned by the tactile tomography system: cleanly resolved plus-sign with step-like vertical extrusion on a red substrate (rainbow Z scale).]

Fig. 9. Theoretical validity of the tactile tomography. (a) Symbol “+” model. (b) Symbol “+” model filled with AB glues. (c) Simulation results of the mechanical process of the probe pressing the sample. (d) Simulation results of the sample scanned by the probe tip with 676 scanning points. (e) Simulated displacement field of the sample under probe loading. (f) Experimental result of the sample scanned by the tactile tomography system.

unable to  in the information of the sample with a thicker soft surface layer if the applied force is small. As shown in Fig. 8a, only the top two stages can be reconstructed and distinguished as the tactile tomography system pressed the sample with a pressure of 0.5 MPa. As the applied force increases, the deeper stage can be reconstructed and distinguished, such as the top 4 stages for the pressure of 1.5 MPa (Fig. 8b), the top 6 stages for the pressure of 2.5 MPa (Fig. 8c), and all stage for the pressure of 4 MPa (Fig. 8d). Therefore, the tactile tomography system can detect and reconstruct the sample layer-by-layer by setting a series of threshold corresponding to increasing pressures. For example, nine thresholds of 0, 1, 2, 3, 4, 5, 6, and 7 that corresponding to the pressures of 0.5, 0.75, 1.10, 1.50, 1.85, 2.50, 4.00, and 5.00 MPa were set, and the reconstructed images can be displayed layer-by-layer in the host computer (as shown in Movie S1). These results proved that this imaging system based on the detection of tactile softness of materials can image an object by tomography. Moreover, Fig. 8d also shows the maximum depth of 6 mm, indicating that the tactile tomography system can identify an object buried 6 mm below the surface. The tomography methodology employed in this tactile tomography system is based on mechanical principles and the novel parameter of “tactile softness of materials,” which simulates human haptic perception by integrating the effects of sample size, thickness, and applied force. The system utilizes a closed-loop feedback control system to drive the probe module to scan the sample in the X-Y plane with a preset scanning step and speed. During scanning, the -axis rodless cylinder applies a series of increasing pressures to the sample through the probe. As the probe interacts with the sample, the pressure sensor detects force changes, and the grating sensor records the corresponding compression depth (Δd) at each X-Y position. This compression depth, combined with the applied force (F) and the contact area (S) between the tip and the sample, defines the tactile softness ($\text{Softness} = \Delta d \cdot S/F$),

which characterizes the material’s softness at that location. By collecting the projection data of positional information (x, y, z) corresponding to each pressure threshold, the system reconstructs layer-by-layer 3D images of the internal structure.

4. Results and discussion

4.1. Theoretical validity of the tactile tomography

To demonstrate the theoretical validity of the tactile tomography, we performed finite element analysis (FEA) to simulate the mechanical interaction between the probe tip and samples with internal features. As shown in Fig. 9a, a model with a feature of a symbol “+” was fabricated by 3D printing, and this model was then filled with a soft surface layer of AB glue (Fig. 9b). A 3D FEA model was built to simulate the mechanical process of the probe pressing the sample (Fig. 9c), the sample undergoes

Fig. 10. Scanning resolution of the tactile tomography system.

- (a) A model featuring topographic lines with line spacings of 2, 1.5, 1, 0.7, 0.5, and 0.3 mm (all lines 2 mm wide, 8 mm long), 3D printed.
- (b) Topographic-lines model filled with a soft surface layer of AB glue (photograph).
- (c) 3D imaging of compression-depth distribution when the tactile tomography system scanned (b). Lines above 0.5 mm spacing remain cleanly distinguishable in the reconstructed depth map; lines at 0.3 mm spacing blur together. [X 0–70 mm, Y 0–20 mm, Z color scale 0.1–5.2 mm.]

Fig. 10. Scanning resolution of the tactile tomography system. (a) A model featuring topographic lines. (b) Topographic lines model filled with a soft surface layer of AB glue. (c) 3D imaging of compression depth distribution when the tactile tomography system scanned the topographic lines model with a soft surface layer.

Fig. 11. Scanning accuracy of the tactile tomography system.

- (a) A gauge block with a length of 30 mm, a width of 9 mm, and a height of 3 mm.
- (b) Gauge block filled with a soft surface layer of AB glue; a 20 mm $\times$ 20 mm scanning area is marked on the top surface.
- (c) 3D imaging of compression-depth distribution when the tactile tomography system scanned (b). The gauge block reads as a flat plateau at $Z \approx 0.1$ mm with a cliff down to $Z \approx 2.4$ mm at the edges.
- (d) Scanning-accuracy distribution corresponding to the compression-depth scan, color-coded 98.95% (lightest) to 100.00% (darkest) across the 20 $\times$ 20 mm scan area.]

Fig. 11. Scanning accuracy of the tactile tomography system. (a) A gauge block with a length of 30 mm, a width of 9 mm, and a height of 3 mm. (b) A gauge block filled with a soft surface layer of AB glue. (c) 3D imaging of compression depth distribution when the tactile tomography system scanned the gauge block with a soft surface layer. (d) Scanning accuracy distribution corresponding to compression depth when the tactile tomography system scanned the gauge block with a soft surface layer.

compression and deformation to a certain displacement as the probe tip presses down (Movie S2). When the sample was scanned by the probe tip with 676 scanning points under a constant pressure of 4 MPa, as shown in Fig. 9d and Movie S3, the “+” shaped region exhibit distinct stress responses compared to the surrounding matrix, and the displacement value of these regions are less than that of the surrounding matrix, reflecting mechanical interaction differences induced by the internal structure. Fig. 9e presents the simulated displacement field of the sample under probe loading. According to the color-code displacement vectors, the “+” shaped region has lower displacement magnitudes than the surrounding matrix, which aligns with the tactile tomography principle that

internal structures alter the effective thickness of the soft surface layer, resulting in different distribution of the tactile softness across the samples. In addition, this simulated result is consistent with the experimental result (as shown in Fig. 9f), confirming that the tactile tomography can effectively translate mechanical interaction differences into detectable internal structure maps.

4.2. Scanning resolution of the tactile tomography system

Scanning resolution is the key performance parameter in determining the detailed quality of scanned imaging. In this article, the x stage, y stage, and z stage of the tactile tomography system can achieve a minimum step displacement of 10, 10, and 0.1 μm , respectively, which ensures the system has a high scanning resolution. To investigate the scanning resolution of the tactile tomography system, a model featuring topographic lines was fabricated by 3D printing. As shown in Fig. 10a, these lines have a length of 8 mm and a width of 2 mm in the x and y directions, and are interspaced by 2.0, 1.5, 1.0, 0.7, 0.5, and 0.3 mm, respectively. The model was then filled with a soft surface layer of AB glue (Fig. 10b), and scanned by the tactile tomography system under a constant pressure of 4 MPa. The scanning result is shown in Fig. 10c, where the distances between the topographic lines in the x and y directions above 0.5 mm can be distinguished, meaning that the tactile tomography system possesses a minimum scanning resolution of 0.5 mm in both the x and y directions. This scanning resolution is typically limited by the diameter of the tip, which is also 0.5 mm. Therefore, the scanning resolution in the x and y directions can be improved by reducing the diameter of the tip. But there is a trade-off between the scanning resolution and the diameter of the tip, because a tip with a very small diameter will pierce the soft surface layer easily. The scanning resolution in the z direction is unaffected by the diameter of the tip, and mainly depends on the minimum step displacement. Thus, the tactile tomography system can achieve a high scanning resolution of 0.1 μm in the z direction, allowing the tactile tomography system to recognize an object in more detail.

4.3. Scanning accuracy of the tactile tomography system

Scanning accuracy is another key performance parameter for the tactile tomography system, referring to its reliability and repeatability.

Fig. 12. Scanning speed and scanning step of the tactile tomography system.

(a)–(c) A spherical-cap model: (a) cross-section with 4 mm AB-glue layer atop ABS substrate; (b) 3D CAD view; (c) photograph of the filled sample.

(d)–(i) 3D imaging of compression-depth distribution when the tactile tomography system scanned the spherical cap under nine speed/step combinations: (d) 0.6 mm/s & 0.5 mm, (e) 1.2 mm/s & 0.5 mm, (f) 1.8 mm/s & 0.5 mm, (g) 2.4 mm/s & 0.5 mm, (h) 2.4 mm/s & 1.0 mm, (i) 2.4 mm/s & 2.0 mm. Each panel renders a rainbow-colored 3D dome on a red substrate; Z scale 0–3.6 mm. Panels (d)–(g) preserve the smooth cap; (h) shows slight facet coarsening; (i) (2 mm step) visibly fails to reproduce the cap shape.]

Fig. 12. Scanning speed and scanning step of the tactile tomography system. (a), (b) and (c) A spherical cap model filled with a soft surface layer of AB glue. 3D imaging of compression depth distribution when the tactile tomography system scanned the spherical cap model with a soft surface layer under scanning speeds and scanning steps of (d) 0.6 mm/s & 0.5 mm, (e) 1.2 mm/s & 0.5 mm, (f) 1.8 mm/s & 0.5 mm, (g) 2.4 mm/s & 0.5 mm, (h) 2.4 mm/s & 1.0 mm, and (i) 2.4 mm/s & 2.0 mm, respectively.

A gauge block with a length of 30 mm, a width of 9 mm, and a height of 3 mm was filled with a soft surface layer of AB glue to investigate the scanning accuracy of the tactile tomography system (Fig. 11a). The scanning area is $20 \times 20 \text{ mm}^2$ (Fig. 11b), and the scanning step is 0.5 mm, resulting in a total number of scanning points of 1600. As shown in Fig. 11c, the scanning points corresponding to the surface of the gauge block have a relatively uniform compression depth. The compression depth of the scanning points in other regions presents a slight difference due to the incomplete smoothing of the bottom, which was prepared by 3D printing with ABS. As a result, a scanning accuracy of over 98.9% is distributed throughout the scanning area (Fig. 11d), and the scanning area corresponding to the surface of the gauge block shows a high scanning accuracy of over 99.3%, demonstrating that the tactile tomography system has the ability to scan every point accurately and stably.

4.4. Scanning speed and scanning step of the tactile tomography system

There is a trade-off between the scanning efficiency and imaging quality for scanning imaging, in which the scanning efficiency is determined by the scanning speed and scanning step. As shown in Fig. 12a, 12b, and 12c, a spherical cap model was filled with a soft surface layer of AB glue to investigate the scanning efficiency and imaging quality of the tactile tomography system. The spherical cap model was scanned by the tactile tomography system with scanning speeds and scanning steps of 0.6 mm/s & 0.5 mm, 1.2 mm/s & 0.5 mm, 1.8 mm/s & 0.5 mm, 2.4 mm/s & 0.5 mm, 2.4 mm/s & 1.0 mm, and 2.4 mm/s & 2.0 mm, respectively. The reconstructed 3D images corresponding to the scanning speeds of 0.6, 1.2, 1.8, and 2.4 mm/s are shown in Fig. 12d, e, f, and g, respectively, and all these 3D images can reproduce the spherical cap model. The curved shape of

the 3D images remained highly smooth as the scanning speed increased to 2.4 mm/s, demonstrating that the increase of scanning speed will not greatly degrade the imaging quality. Fig. 12g, h, and i show the reconstructed 3D images corresponding to the scanning speed and scanning step of 2.4 mm/s & 0.5 mm, 2.4 mm/s & 1.0 mm, and 2.4 mm/s & 2.0 mm, respectively. There is a relative decrease in the smoothness of the curved shape of the 3D images as the scanning step increased, and the 3D image could even not reproduce the spherical cap model well when the scanning step increased to 2 mm. The

Fig. 13. Effect of materials properties on the tactile tomography system.

(a) Schematic: spherical-cap model filled with AB glues of different Shore hardnesses (5, 10, 15, 30 HC) over an ABS substrate.

(b) 3D imaging of the spherical cap with a 5 HC soft surface layer under a constant pressure of 4 MPa.

(c) 10 HC, $P = 4$ MPa. (d) 10 HC, $P = 6$ MPa.

(e) 15 HC, $P = 4$ MPa. (f) 15 HC, $P = 8$ MPa.

(g) 30 HC, $P = 4$ MPa. (h) 30 HC, $P = 13$ MPa.

Each panel renders a rainbow-colored 3D dome; Z scale 0–3.6 mm. Harder surface layers at fixed 4 MPa give only partial cap reconstructions; raising the probe pressure recovers the full cap.]

Fig. 13. Effect of materials properties on the tactile tomography system. (a) A spherical cap model filled with different Shore hardness AB glues. (b) 3D imaging of the spherical cap model with a 5 HC soft surface layer scanned by the system under a constant pressure of 4 MPa. (c) 3D imaging of the spherical cap model with a 10 HC soft surface layer scanned by the system under a constant pressure of 4 MPa. (d) 3D imaging of the spherical cap model with a 10 HC soft surface layer scanned by the system under a constant pressure of 6 MPa. (e) 3D imaging of the spherical cap model with a 15 HC soft surface layer scanned by the system under a constant pressure of 4 MPa. (f) 3D imaging of the spherical cap model with a 15 HC soft surface layer scanned by the system under a constant pressure of 8 MPa. (g) 3D imaging of the spherical cap model with a 30 HC soft surface layer scanned by the system under a constant pressure of 4 MPa. (h) 3D imaging of the spherical cap model with a 30 HC soft surface layer scanned by the system under a constant pressure of 13 MPa.

Fig. 14. Effect of materials types on the tactile tomography system.

(a)–(c) Schematic cross-sections of the spherical-cap model filled with three different soft-layer materials over the ABS substrate: (a) AB glue, (b) PDMS, (c) Ecoflex.

(d)–(f) 3D imaging of compression-depth distribution when the tactile tomography system scanned the spherical cap model with a soft surface layer of (d) AB glue, (e) PDMS, and (f) Ecoflex. Each panel shows a clean rainbow-colored 3D dome on red substrate (Z 0–3.6 mm), demonstrating that the system recovers the cap geometry across all three materials.]

Fig. 14. Effect of materials types on the tactile tomography system. A spherical cap model filled with a soft surface layer of AB glue (a), PDMS (b), and Ecoflex (c), respectively. 3D imaging of compression depth distribution when the tactile tomography system scanned the spherical cap model with a soft surface layer of AB glue (d), PDMS (e), and Ecoflex (f), respectively.

reason for this phenomenon is that the increase in the scanning step will result in a significant reduction in the number of scanning points, such as when the number of scanning points is 1600, 800, and 400 when the scanning step is 0.5, 1.0, and 2.0 mm, respectively. The low number of scanning points make the projection data insufficient, which causes poor quality of the reconstructed 3D images. Therefore, increasing the scanning speed is one effective strategy to achieve both high scanning efficiency and imaging quality for the tactile tomography system.

Fig. 15. Nondestructive testing of encapsulated flexible printed circuit board.

- (a) A flexible printed circuit board (photograph of the bare FPCB with a serpentine copper trace pattern, 15 mm scale).
- (b) Flexible printed circuit board encapsulated with a soft surface layer (photograph of the black encapsulated block).
- (c) Scanning area of the encapsulated FPCB: 28 mm $\times$ 10 mm region with a 7.5 mm inner scale bar, showing the trace pattern through the encapsulation.
- (d) Same as (c) with three injected defects: gap, scratch, cut (marked by pink dashed boxes).
- (e) 3D imaging of compression-depth distribution when the tactile tomography system scanned the encapsulated FPCB (c): clean recovery of the serpentine trace pattern as a rainbow-colored embossed surface (Z 0–0.5 mm).
- (f) Same as (e) for the FPCB with three defects (d): the scan shows trace continuity interruptions at the pink-boxed defect sites.
- (g) Ultrasonic C-scan image of the encapsulated FPCB (c): blurry greenish map with serpentine outline barely resolved.
- (h) Ultrasonic C-scan image of the FPCB with three defects (d): similarly low-contrast map with the defect sites indicated but less distinguishable than in the tactile-tomography scan.]

Fig. 15. Nondestructive testing of encapsulated flexible printed circuit board. (a) A flexible printed circuit board. (b) A flexible printed circuit board encapsulated with a soft surface layer. (c) Scanning area of the encapsulated flexible printed circuit board. (d) Scanning area of the encapsulated flexible printed circuit board with three defects. (e) 3D imaging of compression depth distribution when the tactile tomography system scanned the encapsulated flexible printed circuit board. (f) 3D imaging of compression depth distribution when the tactile tomography system scanned the encapsulated flexible printed circuit board with three defects. (g) Ultrasonic C-scan image of the encapsulated flexible printed circuit board. (h) Ultrasonic C-scan image of the encapsulated flexible printed circuit board with three defects.

4.5. Effect of materials properties and materials types on the tactile tomography system

The property of the surface layer materials, especially hardness, can have an impact on the imaging quality of the tactile tomography system. As shown in Fig. 13a, the soft surface layer of AB glue was prepared with different Shore hardness values of 5, 10, 15, and 30 HC, respectively. When the samples were scanned by the tactile tomography system with a scanning speed and scanning step of 2.4 mm/s & 0.5 mm under a constant pressure of 4 MPa (Fig. 13b, c, e, and g), the area where the topographical profile of the spherical cap model can be reconstructed gradually decreased from the bottom to the top as Shore hardness of the soft surface layer increased. Even only the top part of the spherical cap model can be reconstructed when the Shore hardness of the soft surface layer is 30 HC (Fig. 13g). The reason is that the increase in hardness of the surface layer will reduce the compression depth of the probe tip when the samples are scanned by the tactile tomography system under the same constant pressure, resulting in the inability of this system to detect and reconstruct the complete internal structure of the samples. Significantly, the topographical profile of the spherical cap model can be completely reconstructed and maintain excellent imaging quality when the probe pressure is increased for the samples with harder AB glue layer ($P = 6$ MPa for 10

HC (Fig. 13d), $P = 8$ MPa for 15 HC (Fig. 13f), and $P = 13$ MPa for 30 HC (Fig. 13h)). Although the increase of hardness of the surface layer, the tactile tomography system can still maintain a good imaging quality by increasing the probe pressure. The tactile tomography system can adapt to the change in the mechanical properties of AB glue with different Shore hardness by adjusting the probe pressure. Therefore, the tactile tomography system demonstrates excellent adaptability to the surface layer materials with different Shore hardness.

In addition to the AB glue, the other two commonly used soft materials, PDMS and Ecoflex, were also used as the surface layer to further investigate the versatility of the tactile tomography system in characterizing samples with different materials composition. As shown in Fig. 14a, b, and c, the spherical cap model filled with a soft surface layer of AB glue, PDMS, and Ecoflex, respectively. These samples were

Fig. 16. Fatigue testing of encapsulated flexible printed circuit board.

- (a) Compression depth of the probe tip at the test point of the encapsulated FPCB under 300 cycles of scanning: depth vs. cycle number 0–300, applied pressure 4 MPa, line held at ≈ 0.18 mm across all cycles (negligible drift).
- (b) Microscope image of the test point of the encapsulated FPCB before scanning: intact orange serpentine trace with a smooth encapsulated surface.
- (c) Microscope image after 300 cycles of scanning: an almost indiscernible circular indentation mark on the AB-glue encapsulation layer, no crack or puncture.
- (d) Microscope image of the FPCB itself after 300 cycles of scanning: the underlying circuit remains intact with no visible deformation.]

Fig. 16. Fatigue testing of encapsulated flexible printed circuit board. (a) Compression depth of the probe tip at the test point of the encapsulated FPCB under 300 cycles of scanning. (b) Microscope image of the test point of the encapsulated FPCB before scanning. (c) Microscope image of the test point of the encapsulated FPCB after 300 cycles scanning. (d) Microscope image of the test point of the FPCB after 300 cycles scanning.

scanned by the tactile tomography system with a scanning speed and scanning step of 2.4 mm/s & 0.5 mm under a constant pressure of 4 MPa. Fig. 14d, e, and f presents the reconstructed 3D images corresponding to the soft surface layer of AB glue, PDMS, and Ecoflex, respectively. All these 3D images successfully reconstructed the topographical profile of the spherical cap model beneath the soft surface layer of AB glue, PDMS, and Ecoflex, indicating that the tactile tomography system has excellent universality for different materials.

4.6. Application in nondestructive testing of encapsulated flexible printed circuit boards (FPCBs)

Defect Detection of encapsulated flexible printed circuit boards (FPCBs) is one of the most promising applications for the tactile tomography system. FPCBs as a popular substrate has been widely used in flexible wearable devices [39]. In practice, flexible wearable devices are usually encapsulated with a soft layer to protect them from the outside environment without sacrificing their flexibility [40]. However, the current research mainly focuses on the defect detection of FPCBs without packaging. Therefore, nondestructive testing of encapsulated FPCBs is still a challenge. As shown in Fig. 15a, an FPCB sample consisting of PI and copper lines was purchased, and it was then encapsulated with a soft layer of AB glue, which was dyed black to prevent detection by the human eye and optical microscope (Fig. 15b). The encapsulated FPCBs sample without any defect was scanned by the tactile tomography system with a scanning speed and scanning step of 0.6 mm/s & 0.5 mm under an effective scanning range of $28 \times 10 \text{ mm}^2$ (Fig. 15c). The reconstructed 3D image is shown in Fig. 15e, in which the layout of the copper lines was completely reproduced, demonstrating that the tactile tomography system can detect and image the internal structure of flexible wearable devices without peeling off the encapsulated layer. To investigate the defect detection

of the encapsulated FPCB, three defects of open the copper lines on the FPCB were made (as shown in Fig. 15d). These defects can be found easily on the reconstructed 3D image in Fig. 15f, meaning that the tactile tomography system possesses the capability for defect detection in encapsulated FPCBs. For comparison, ultrasonic imaging (Ultrasonic detection system, Feli 650) was also employed to inspect the same encapsulated FPCB sample, the corresponding ultrasonic C-scan images are presented in Fig. 15g and h. In Fig. 15g, the ultrasonic image of the intact encapsulated FPCB shows a relatively blurred circuit pattern, and in Fig. 15h, when defects exist, the ultrasonic image fails to clearly delineate the defects. This is because ultrasonic imaging is susceptible to the influence of the encapsulation layer. The acoustic properties of the encapsulation material differ from those of the FPCB, causing scattering and attenuation of ultrasonic waves, which degrades the imaging quality and makes it difficult to accurately identify internal defects. This limitation arises from the acoustic impedance mismatch and the wave-damping nature of the soft

polymeric materials typically used for encapsulation. In stark contrast, the proposed tactile tomography system, which relies on mechanical sensing rather than acoustic wave propagation, is not susceptible to such limitations. Its imaging efficacy is inherently based on the local mechanical interaction (tactile softness) between the probe and the sample, which effectively bypasses the interfering effects of the encapsulation layer. This key advantage allows the tactile tomography system to provide superior imaging performance for soft-encapsulated structures, highlighting its unique potential for nondestructive testing in applications where traditional ultrasonic methods may fail. To demonstrate the non-destructiveness of the tactile tomography system, a targeted fatigue test was conducted on the encapsulated FPCB sample. A fixed point on the encapsulated FPCB was continuously scanned 300 times using the tactile tomography system under the test conditions of an applied pressure of 4 MPa, a scanning speed and scanning step of 0.6 mm/s & 0.5 mm. Under 300 cycles of constant-pressure cyclic compression, the compression depth of the probe tip at the test point remained stable (Fig. 16a), indicating that no stress distortion occurred in the interaction between the probe and the encapsulated FPCB during repeated loading, and thus the FPCB did not suffer cumulative mechanical damage. Fig. 16b shows the microscope image of the test point of the encapsulated FPCB before scanning. After 300 cycles of scanning, the post-scanning image is shown in Fig. 16c, only an almost indiscernible indentation mark was left on the surface of AB glue encapsulation layer, it neither cracked nor was punctured, and the underlying FPCB remained intact with no visible deformation (Fig. 16d). These results fully demonstrate that the tactile tomography system does not exacerbate initial or cumulative damage to the encapsulated FPCB.

5. Conclusion

In this work, a novel tactile tomography system based on mechanical principles with internal 3D imaging capabilities was developed. This tactile tomography system can recognize the softness of AB glues with Shore hardnesses of 3, 7.5, 12.5, 17.5, 25, 30, 35, 50, and 60 HC. For samples with dimensions less than 12 mm in size and 6 mm in thickness, a novel parameter of “tactile softness of materials” was introduced, enabling the tactile tomography system to perform internal 3D imaging with a maximum detection depth of 6 mm. The tactile tomography system features a high scanning resolution of 0.5 mm in both the x and y directions, and 0.1 μm in the z direction. The tactile tomography system possesses a scanning accuracy of over 98.9%, and an optimal scanning speed and scanning step of 2.4 mm/s & 0.5 mm. This system is capable of imaging the internal structure and identifying defects in encapsulated FPCBs, demonstrating significant potential for nondestructive testing of flexible wearable devices.

CRedit authorship contribution statement

Zhiming Chen: Writing – review & editing, Visualization, Methodology, Investigation. **Longxin Qian:** Writing – original draft, Software, Data curation. **Weihao Liang:** Writing – original draft, Formal analysis, Conceptualization. **Kaibang Zhang:** Software, Data curation. **Zichen Liu:** Investigation, Data curation. **Fengming Hu:** Investigation, Data curation. **Chaobin Liu:** Investigation, Data curation. **Kelan Lu:** Investigation, Data curation. **Jingcheng Huang:** Investigation. **Haiquan Li:** Methodology. **Jinxiu Wen:** Visualization, Methodology, Investigation. **Bingpu Zhou:** Methodology, Investigation. **Jianyi Luo:** Writing – review & editing, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.measurement.2026.120300>.

Data availability

Data will be made available on request.

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